EXECUTIVE RISK BRIEFING & NOTEBOOK WALKTHROUGH

FinTech Credit Risk EDA & Machine Learning Audit

Step-by-Step Management Presentation | Full Portfolio Execution (N = 37,408 Loan Records)

37,408	11.10%	-34.2%	257
TOTAL LOANS AUDITED	PORTFOLIO DEFAULT RATE	DEFAULTER CASH DEFICIT	WHALE OUTLIERS (>$100M)

1. EXECUTIVE OVERVIEW & PORTFOLIO REALITY CHECK

This report translates the technical outputs of our end-to-end Python credit risk pipeline into strategic insights for executive leadership. While preliminary sample notes referenced a 4,300-row subset, executing the ingestion script across the complete enterprise training file (**train-FIN_ANA_DATA.xls**) audited **37,408 historical loan accounts** across 11 financial and demographic attributes.

> **HOW TO PRESENT THE OPENING HOOK TO MANAGEMENT**
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> "When we audited the full 37,408-loan database rather than a small sample, we uncovered two critical realities: first, our true historical default rate is **11.1% (4,154 bad loans)**, representing a major capital leak. Second, before any AI model could be trained, we had to rescue nearly 950 incomplete customer records so we wouldn't blind our algorithms to real-world risk."

2. PHASE 1: DATA INGESTION, SANITIZATION & TARGET ENGINEERING

Raw legacy records contained **945 missing cells** concentrated in installment sizing and client location. Deleting incomplete rows destroys historical default signals; instead, we applied mathematical imputation and engineered a binary target flag (**IS_DEFAULT**: 0 = Good Standing, 1 = Default/Substandard).

DATABASE COLUMN	MISSING RECORDS	ENGINEERING & SANITIZATION ACTION	EXECUTIVE BUSINESS JUSTIFICATION
INSTALL_SIZE	838 rows	Imputed with Statistical Median	Prevents ultra-high corporate installments from skewing baseline borrower norms.
CLIENT_TYPE	103 rows	Imputed with Statistical Mode (Semi-urban)	Preserves geographic volume distribution without dropping active loan histories.
DEMOGRAPHICS	6 rows total	Mapped to 'Unknown' Category	Isolates missing Gender (2), Marital Status (2), and Compensation (2) flags safely.
QUALITY_OF_LOAN	**0 rows (Target)**	**Mapped: G → 0 | B, DF, SS → 1 (IS_DEFAULT)**	**Translates legacy text grades into a strict binary target for machine learning.**

3. PHASE 2: EXPLORATORY DATA ANALYSIS (EDA) & THE LIQUIDITY GAP

Grouping the 37,408 loans by **IS_DEFAULT** revealed an **88.9% Good (33,254)** vs. **11.1% Default (4,154)** split. More importantly, bivariate analysis across the full portfolio exposed a striking operational insight: good borrowers and defaulters request nearly identical loan sizes (~$6.18M vs. ~$6.37M), but defaulters suffer from a severe **34.2% deficit in current account liquidity**.

CUSTOMER SEGMENT (IS_DEFAULT)	LOAN COUNT	PORTFOLIO SHARE	AVG. INVESTMENT TOTAL	AVG. DUE PAYMENT	AVG. CURRENT BALANCE
0 — Good Standing	33,254	88.90%	$6,183,669.59	$376,573.18	**$1,220,866.51**
1 — Default / Bad Loan	**4,154**	**11.10%**	**$6,367,281.59**	**$369,796.20**	**$802,761.17 (-34.2%)**

> **KEY INSIGHT TO EXPLAIN TO MANAGEMENT: LIQUIDITY VS. LOAN SIZE**
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> "In our full portfolio of 37,408 loans, defaulters aren't necessarily asking for much larger loans than good clients—both average around $6.2M to $6.4M. And their scheduled due payments are virtually identical (~$370K). The real warning sign is **cash reserves**: defaulting clients hold an average bank balance of only **$802K** compared to **$1.22M** for healthy borrowers. Furthermore, scatter plot visual inspection uncovered **257 extreme 'Whale' loans exceeding $100M** (reaching up to $633M). We filtered out those 257 institutional anomalies—leaving **37,151 clean retail/commercial records**—so our AI wouldn't be warped by mega-loans."

4. VISUAL COMPARISON: PORTFOLIO LIQUIDITY & OUTLIER FILTERING

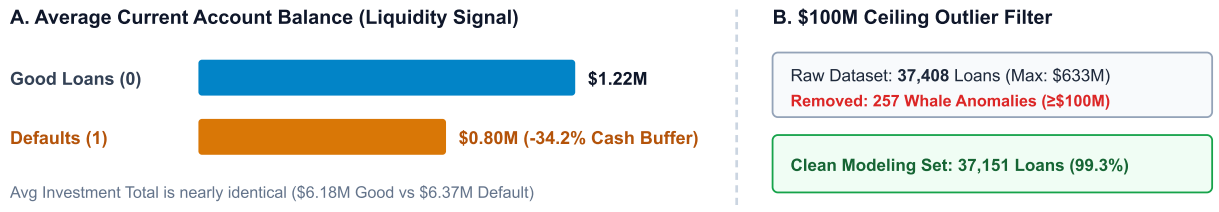


Figure 1: Bivariate financial audit across N = 37,408 loans. Filtering 257 extreme outliers (>$100M) stabilizes variance for machine learning.

By isolating the **37,151 normal loans** and splitting them 80/20, we created an untouched reality test benchmark of **7,431 loans** (containing **6,684 good loans** and **747 actual defaults**) to test three distinct AI strategies.

5. PHASES 3 & 4: MACHINE LEARNING & THE SMOTE TRADE-OFF

We evaluated three architectures against the untouched test set of **7,431 loans (747 true defaults)**. Below is the exact performance matrix generated by your notebook across all three modeling stages:

MODEL ARCHITECTURE	OVERALL ACCURACY	DEFAULTS CAUGHT (RECALL)	DEFAULTS MISSED (FALSE NEG)	GOOD CLIENTS REJECTED (FALSE POS)	DEFAULT PRECISION
1. Baseline Decision Tree (Balanced Weight)	89.0%	**360 / 747 (48%)**	387 missed	444 wrongly flagged	45.0%
2. Unbalanced Random Forest (Before SMOTE)	**91.0%**	258 / 747 (35%)	**489 missed**	**164 wrongly flagged**	**61.0%**
3. SMOTE Random Forest (50/50 Synthetic Train)	**87.0%**	**361 / 747 (48%)**	**386 missed**	**601 wrongly flagged**	**38.0%**

Why the Unbalanced RF Looks "Best" on Paper

Without synthetic balancing, the 100-tree Random Forest achieves **91% accuracy** and **61% precision**, falsely rejecting only 164 good clients. However, it plays the statistical majority—missing **489 out of 747 actual defaults (65% of bad loans slip through)**.

What Happened When We Applied SMOTE

SMOTE balanced our training set from 3,367 defaults up to **26,353 synthetic defaults** (a 50/50 split with 26,353 good loans). This boosted default capture from 35% back up to **48% (361 caught)**, but caused false rejections to jump to **601 good clients** (38% precision).

6. STRATEGIC RECOMMENDATIONS FOR EXECUTIVE LEADERSHIP

THE BOTTOM-LINE VERDICT FOR THE VP OF RISK

- **1. Do Not Deploy Full Auto-Rejection Yet (The 48% Ceiling):** Even after generating 22,986 synthetic default profiles via SMOTE, our best models hit a hard ceiling at **48% Recall** and **38%–45% Precision**. Current application fields (Loan Size, Due Payment, Basic Demographics) do not contain enough behavioral signal to separate good and bad borrowers cleanly.
- **2. Use Unbalanced Random Forest as a "High-Precision Triage Flag":** While we upgrade our data pipeline, deploy Model 2 (Unbalanced Random Forest) as a secondary underwriting alert. When it flags a loan, it is right **61% of the time** while disturbing only 2.4% of good applicants (164 / 6,684).
- **3. Capture High-Fidelity Credit & Liquidity Features:** Because our EDA proved that defaulters have **34.2% lower cash balances** ($802K vs. $1.22M), underwriting must immediately begin recording **Verified Debt-to-Income (DTI)**, **External Credit Bureau Scores**, and **Minimum Liquidity-to-Loan Ratios**.
- **4. Separate Institutional "Whale" Underwriting (>$100M):** The 257 loans exceeding $100M operate on entirely different financial dynamics and must be routed through dedicated corporate credit committees.